Here we consider small data

QM2D With Dropped Duplicates (**42** train (**0.01**), 841 test (0.2), overall: 4201 planar molecules). Core, Dirac, valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats

2025_02_03.ipynb	Scattering rotation-invariant				Vanilla Scattering				Coulomb				CNN			
	scat8_rotinv_qm2d GP(RBF)	scat8_rotinv_qm2d GP(RBF,ARD)	scat8_rotinv_qm2d GP(Matern32)	scat8_rotinv_qm2d GP(Matern32,ARD)	scat8_qm2d GP(RBF)	scat8_qm2d GP(RBF,ARD)	scat8_qm2d GP(Matern32)	scat8_qm2d GP(Matern32,ARD)	qm2d_coulomb GP(RBF)	qm2d_coulomb GP(RBF,ARD)	qm2d_coulomb GP(Matern32)	qm2d_coulomb GP(Matern32,ARD)	qm2d_cnn GP(RBF)	qm2d_cnn GP(RBF,ARD)	qm2d_cnn GP(Matern32)	qm2d_cnn GP(Matern32,ARD)
RMSE	0.66±0.11 (0 Nones, min: 0.55, max: 0.85)	0.59±0.05 (0 Nones, min: 0.53, max: 0.66)	0.91±0.35 (0 Nones, min: 0.52, max: 1.45)	**0.6±0.06 (0 Nones, min: 0.51, max: 0.69)**	1.02±0.17 (0 Nones, min: 0.78, max: 1.32)	0.73±0.16 (0 Nones, min: 0.44, max: 0.88)	0.76±0.1 (0 Nones, min: 0.61, max: 0.88)	0.73±0.08 (0 Nones, min: 0.58, max: 0.82)	0.88±0.07 (0 Nones, min: 0.8, max: 0.99)	0.9±0.15 (0 Nones, min: 0.72, max: 1.17)	0.77±0.11 (0 Nones, min: 0.56, max: 0.89)	0.85±0.13 (0 Nones, min: 0.65, max: 1.05)	0.84±0.07 (0 Nones, min: 0.75, max: 0.92)	0.91±0.16 (0 Nones, min: 0.72, max: 1.2)	0.81±0.11 (0 Nones, min: 0.69, max: 1.03)	0.8±0.06 (0 Nones, min: 0.71, max: 0.86)
RMSE (unnormalized)	121.1±20.54 (0 Nones, min: 101.47, max: 159.81)	105.72±5.59 (0 Nones, min: 97.36, max: 112.34)	144.03±30.17 (0 Nones, min: 101.31, max: 185.44)	**102.31±10.65 (0 Nones, min: 88.73, max: 120.75)**	161.26±16.36 (0 Nones, min: 142.07, max: 185.05)	129.2±11.95 (0 Nones, min: 107.95, max: 140.72)	151.74±13.98 (0 Nones, min: 137.66, max: 175.96)	114.33±11.48 (0 Nones, min: 95.4, max: 127.7)	148.75±4.6 (0 Nones, min: 140.14, max: 153.56)	156.94±3.67 (0 Nones, min: 150.96, max: 160.39)	146.38±16.62 (0 Nones, min: 126.3, max: 174.22)	144.38±10.79 (0 Nones, min: 127.47, max: 154.96)	145.42±6.12 (0 Nones, min: 135.23, max: 153.32)	162.04±7.56 (0 Nones, min: 148.69, max: 168.49)	143.19±9.8 (0 Nones, min: 127.07, max: 156.51)	147.77±5.05 (0 Nones, min: 141.52, max: 156.31)
MAE	0.5±0.1 (0 Nones, min: 0.4, max: 0.67)	0.45±0.04 (0 Nones, min: 0.39, max: 0.51)	0.7±0.27 (0 Nones, min: 0.41, max: 1.11)	**0.45±0.03 (0 Nones, min: 0.4, max: 0.5)**	0.78±0.13 (0 Nones, min: 0.59, max: 1.0)	0.58±0.13 (0 Nones, min: 0.35, max: 0.69)	0.57±0.07 (0 Nones, min: 0.48, max: 0.65)	0.55±0.05 (0 Nones, min: 0.46, max: 0.62)	0.67±0.05 (0 Nones, min: 0.61, max: 0.75)	0.68±0.13 (0 Nones, min: 0.5, max: 0.9)	0.57±0.09 (0 Nones, min: 0.44, max: 0.67)	0.63±0.1 (0 Nones, min: 0.47, max: 0.79)	0.64±0.04 (0 Nones, min: 0.57, max: 0.69)	0.71±0.12 (0 Nones, min: 0.56, max: 0.92)	0.62±0.09 (0 Nones, min: 0.54, max: 0.79)	0.63±0.05 (0 Nones, min: 0.56, max: 0.67)
MAE (unnormalized)	92.17±18.25 (0 Nones, min: 73.75, max: 125.13)	79.85±4.71 (0 Nones, min: 73.98, max: 85.82)	110.23±22.48 (0 Nones, min: 80.6, max: 141.99)	**76.91±5.01 (0 Nones, min: 70.75, max: 86.1)**	124.13±12.99 (0 Nones, min: 107.36, max: 143.66)	101.38±9.33 (0 Nones, min: 85.86, max: 111.21)	114.05±8.98 (0 Nones, min: 104.93, max: 130.28)	86.78±7.79 (0 Nones, min: 75.27, max: 96.39)	114.08±4.37 (0 Nones, min: 106.78, max: 119.87)	118.78±4.76 (0 Nones, min: 112.78, max: 126.51)	107.31±6.81 (0 Nones, min: 98.35, max: 118.17)	106.6±8.01 (0 Nones, min: 92.9, max: 116.15)	111.36±6.01 (0 Nones, min: 102.65, max: 121.51)	126.03±5.92 (0 Nones, min: 115.1, max: 131.63)	109.8±7.18 (0 Nones, min: 99.21, max: 120.2)	115.83±3.21 (0 Nones, min: 112.08, max: 120.81)
NLL (full)	**0.42±0.21 (0 Nones, min: 0.15, max: 0.66)**	25.37±12.81 (0 Nones, min: 11.34, max: 48.59)	0.73±0.56 (0 Nones, min: -0.07, max: 1.63)	4.32±1.41 (0 Nones, min: 1.6, max: 5.56)	0.88±0.44 (0 Nones, min: 0.12, max: 1.31)	14.69±5.78 (0 Nones, min: 6.57, max: 21.38)	0.54±0.35 (0 Nones, min: 0.07, max: 1.11)	3.37±0.42 (0 Nones, min: 2.8, max: 3.85)	0.93±0.06 (0 Nones, min: 0.87, max: 1.05)	0.99±0.24 (0 Nones, min: 0.6, max: 1.33)	0.57±0.18 (0 Nones, min: 0.31, max: 0.83)	0.47±0.38 (0 Nones, min: -0.15, max: 0.89)	0.71±0.2 (0 Nones, min: 0.4, max: 0.88)	9.61±5.7 (0 Nones, min: 4.88, max: 20.67)	0.71±0.05 (0 Nones, min: 0.68, max: 0.82)	2.04±0.43 (0 Nones, min: 1.52, max: 2.79)
NLL (diag)	**773.21±158.58 (0 Nones, min: 614.3, max: 1021.37)**	7713.39±3185.25 (0 Nones, min: 4010.32, max: 12795.	1089.17±370.7 (0 Nones, min: 697.51, max: 1676.04)	2683.07±937.03 (0 Nones, min: 1115.96, max: 3855.1	1225.64±177.23 (0 Nones, min: 983.8, max: 1533.44)	6403.59±2142.47 (0 Nones, min: 3977.58, max: 9555.2	947.92±109.78 (0 Nones, min: 780.0, max: 1092.26)	2345.0±407.77 (0 Nones, min: 1827.02, max: 2901.5)	1111.88±98.66 (0 Nones, min: 998.55, max: 1289.04)	1109.48±188.07 (0 Nones, min: 877.92, max: 1444.48)	910.43±123.13 (0 Nones, min: 750.4, max: 1070.96)	1056.33±164.77 (0 Nones, min: 809.21, max: 1284.47)	1023.31±56.07 (0 Nones, min: 941.79, max: 1093.	3936.66±1852.93 (0 Nones, min: 2491.56, max: 74	1004.16±127.93 (0 Nones, min: 870.51, max: 1246.	2092.63±455.85 (0 Nones, min: 1603.75, max: 2956.13)
Time (s)	182.16±95.38 (0 Nones, min: 47.53, max: 322.55)	282.06±99.68 (0 Nones, min: 135.62, max: 412.98)	180.88±92.81 (0 Nones, min: 48.21, max: 311.03)	266.39±90.67 (0 Nones, min: 137.44, max: 392.87)	191.8±98.96 (0 Nones, min: 46.66, max: 326.31)	828.49±108.8 (0 Nones, min: 676.78, max: 983.5)	187.72±96.79 (0 Nones, min: 48.4, max: 316.49)	843.23±95.73 (0 Nones, min: 704.46, max: 970.39)	167.9±87.05 (0 Nones, min: 44.98, max: 292.88)	177.26±88.05 (0 Nones, min: 53.59, max: 302.11)	166.04±87.19 (0 Nones, min: 42.79, max: 289.32)	175.16±85.81 (0 Nones, min: 54.33, max: 296.14)	161.48±84.15 (0 Nones, min: 43.2, max: 281.08)	173.72±85.95 (0 Nones, min: 52.47, max: 295.31)	164.69±85.54 (0 Nones, min: 43.96, max: 285.97)	174.73±86.48 (0 Nones, min: 52.11, max: 297.6)

QM2D With Dropped Duplicates (**84** train (**0.02**), 841 test (0.2), overall: 4201 planar molecules). Core, Dirac, valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats

2025_02_03.ipynb	Scattering rotation-invariant				Vanilla Scattering				Coulomb				CNN			
	scat8_rotinv_qm2d GP(RBF)	scat8_rotinv_qm2d GP(RBF,ARD)	scat8_rotinv_qm2d GP(Matern32)	scat8_rotinv_qm2d GP(Matern32,ARD)	scat8_qm2d GP(RBF)	scat8_qm2d GP(RBF,ARD)	scat8_qm2d GP(Matern32)	scat8_qm2d GP(Matern32,ARD)	qm2d_coulomb GP(RBF)	qm2d_coulomb GP(RBF,ARD)	qm2d_coulomb GP(Matern32)	qm2d_coulomb GP(Matern32,ARD)	qm2d_cnn GP(RBF)	qm2d_cnn GP(RBF,ARD)	qm2d_cnn GP(Matern32)	qm2d_cnn GP(Matern32,ARD)
RMSE	0.46±0.08 (0 Nones, min: 0.32, max: 0.55)	**0.32±0.03 (0 Nones, min: 0.29, max: 0.36)**	0.46±0.03 (0 Nones, min: 0.39, max: 0.49)	0.47±0.08 (0 Nones, min: 0.39, max: 0.62)	0.45±0.12 (0 Nones, min: 0.35, max: 0.68)	0.4±0.11 (0 Nones, min: 0.3, max: 0.58)	0.57±0.12 (0 Nones, min: 0.4, max: 0.76)	0.45±0.07 (0 Nones, min: 0.32, max: 0.53)	0.74±0.07 (0 Nones, min: 0.64, max: 0.85)	0.7±0.09 (0 Nones, min: 0.55, max: 0.81)	0.55±0.02 (0 Nones, min: 0.52, max: 0.58)	0.52±0.05 (0 Nones, min: 0.43, max: 0.57)	0.59±0.04 (0 Nones, min: 0.53, max: 0.65)	0.72±0.1 (0 Nones, min: 0.54, max: 0.85)	0.62±0.08 (0 Nones, min: 0.49, max: 0.71)	0.76±0.08 (0 Nones, min: 0.63, max: 0.84)
RMSE (unnormalized)	83.57±5.82 (0 Nones, min: 72.06, max: 87.69)	**61.93±6.89 (0 Nones, min: 52.83, max: 70.03)**	80.92±3.03 (0 Nones, min: 76.37, max: 84.1)	77.21±12.42 (0 Nones, min: 64.78, max: 98.17)	87.11±12.76 (0 Nones, min: 71.58, max: 109.99)	76.12±8.97 (0 Nones, min: 64.71, max: 87.22)	97.86±9.32 (0 Nones, min: 84.82, max: 113.61)	80.88±9.76 (0 Nones, min: 64.19, max: 90.9)	131.22±5.77 (0 Nones, min: 123.3, max: 139.63)	132.69±9.84 (0 Nones, min: 114.45, max: 144.11)	104.0±4.88 (0 Nones, min: 99.69, max: 111.57)	97.45±6.33 (0 Nones, min: 91.69, max: 106.99)	111.81±6.36 (0 Nones, min: 101.27, max: 119.2)	130.92±11.7 (0 Nones, min: 112.64, max: 147.7)	111.83±2.76 (0 Nones, min: 107.17, max: 114.19)	128.13±5.62 (0 Nones, min: 120.57, max: 136.94)
MAE	0.34±0.05 (0 Nones, min: 0.24, max: 0.4)	**0.24±0.01 (0 Nones, min: 0.22, max: 0.26)**	0.34±0.02 (0 Nones, min: 0.3, max: 0.37)	0.34±0.05 (0 Nones, min: 0.29, max: 0.42)	0.35±0.09 (0 Nones, min: 0.27, max: 0.52)	0.31±0.08 (0 Nones, min: 0.23, max: 0.45)	0.44±0.09 (0 Nones, min: 0.29, max: 0.59)	0.34±0.06 (0 Nones, min: 0.25, max: 0.39)	0.56±0.06 (0 Nones, min: 0.49, max: 0.66)	0.51±0.07 (0 Nones, min: 0.41, max: 0.6)	0.39±0.01 (0 Nones, min: 0.37, max: 0.4)	0.36±0.03 (0 Nones, min: 0.31, max: 0.4)	0.46±0.03 (0 Nones, min: 0.41, max: 0.5)	0.57±0.08 (0 Nones, min: 0.42, max: 0.67)	0.47±0.05 (0 Nones, min: 0.38, max: 0.54)	0.58±0.07 (0 Nones, min: 0.46, max: 0.64)
MAE (unnormalized)	61.72±3.3 (0 Nones, min: 55.26, max: 64.23)	**45.94±4.32 (0 Nones, min: 40.47, max: 51.94)**	60.61±1.76 (0 Nones, min: 58.81, max: 63.42)	55.65±7.07 (0 Nones, min: 47.57, max: 66.27)	66.34±9.55 (0 Nones, min: 55.45, max: 83.97)	59.16±7.14 (0 Nones, min: 50.42, max: 67.16)	75.43±8.23 (0 Nones, min: 62.3, max: 87.96)	60.49±7.73 (0 Nones, min: 49.06, max: 71.23)	98.99±6.04 (0 Nones, min: 90.5, max: 108.67)	96.66±7.02 (0 Nones, min: 85.63, max: 106.44)	72.94±2.68 (0 Nones, min: 69.43, max: 77.21)	67.13±4.0 (0 Nones, min: 63.42, max: 74.7)	85.92±4.71 (0 Nones, min: 78.47, max: 93.3)	102.54±9.88 (0 Nones, min: 87.43, max: 117.03)	86.23±1.95 (0 Nones, min: 84.03, max: 89.33)	97.76±5.92 (0 Nones, min: 87.35, max: 105.33)
NLL (full)	0.03±0.21 (0 Nones, min: -0.34, max: 0.25)	10.65±2.55 (0 Nones, min: 6.35, max: 13.89)	0.05±0.08 (0 Nones, min: -0.07, max: 0.18)	1.35±0.38 (0 Nones, min: 0.72, max: 1.75)	**-0.12±0.23 (0 Nones, min: -0.34, max: 0.2)**	8.35±1.11 (0 Nones, min: 6.91, max: 10.19)	0.25±0.21 (0 Nones, min: -0.08, max: 0.5)	0.96±0.56 (0 Nones, min: 0.0, max: 1.68)	0.73±0.1 (0 Nones, min: 0.63, max: 0.91)	0.62±0.24 (0 Nones, min: 0.33, max: 0.88)	0.06±0.09 (0 Nones, min: -0.04, max: 0.21)	**-0.12±0.08 (0 Nones, min: -0.21, max: 0.02)**	0.44±0.13 (0 Nones, min: 0.33, max: 0.67)	5.37±0.81 (0 Nones, min: 4.4, max: 6.59)	0.46±0.13 (0 Nones, min: 0.23, max: 0.58)	1.26±0.24 (0 Nones, min: 0.91, max: 1.58)
NLL (diag)	**434.75±138.37 (0 Nones, min: 168.23, max: 552.28)**	4322.51±1391.17 (0 Nones, min: 2108.27, max: 5736.0	475.99±65.29 (0 Nones, min: 361.6, max: 547.62)	1410.47±164.54 (0 Nones, min: 1149.57, max: 1647.3	489.06±224.31 (0 Nones, min: 305.65, max: 902.63)	4998.92±998.19 (0 Nones, min: 3801.63, max: 6577.76	694.57±182.67 (0 Nones, min: 381.41, max: 952.5)	1408.53±847.29 (0 Nones, min: 351.39, max: 2945.56)	918.11±103.56 (0 Nones, min: 782.32, max: 1077.76)	813.65±147.66 (0 Nones, min: 642.72, max: 987.32)	580.63±30.42 (0 Nones, min: 552.25, max: 637.23)	476.39±58.12 (0 Nones, min: 384.19, max: 562.72)	752.21±63.47 (0 Nones, min: 655.52, max: 829.35	2638.92±622.62 (0 Nones, min: 1948.76, max: 37	770.01±94.58 (0 Nones, min: 601.42, max: 861.96)	1358.34±189.03 (0 Nones, min: 1098.59, max: 1587.41)
Time (s)	184.06±94.79 (0 Nones, min: 46.0, max: 317.98)	263.34±87.08 (0 Nones, min: 140.42, max: 389.22)	181.63±92.09 (0 Nones, min: 45.51, max: 301.34)	270.1±93.35 (0 Nones, min: 139.66, max: 403.24)	186.36±95.58 (0 Nones, min: 47.8, max: 320.29)	820.38±105.24 (0 Nones, min: 662.21, max: 960.14)	186.99±96.46 (0 Nones, min: 47.6, max: 324.96)	814.63±107.99 (0 Nones, min: 657.4, max: 967.63)	174.97±90.35 (0 Nones, min: 43.21, max: 296.76)	178.43±87.6 (0 Nones, min: 55.37, max: 303.14)	167.57±87.0 (0 Nones, min: 44.32, max: 290.97)	176.39±86.2 (0 Nones, min: 55.03, max: 298.06)	161.64±83.95 (0 Nones, min: 42.71, max: 280.55)	175.75±86.87 (0 Nones, min: 53.79, max: 300.27)	165.74±86.38 (0 Nones, min: 44.0, max: 288.98)	175.57±85.44 (0 Nones, min: 54.57, max: 295.51)

QM2D With Dropped Duplicates (**126** train (**0.03**), 841 test (0.2), overall: 4201 planar molecules). Core, Dirac, valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats

2025_02_03.ipynb	Scattering rotation-invariant				Vanilla Scattering				Coulomb				CNN			
	scat8_rotinv_qm2d GP(RBF)	scat8_rotinv_qm2d GP(RBF,ARD)	scat8_rotinv_qm2d GP(Matern32)	scat8_rotinv_qm2d GP(Matern32,ARD)	scat8_qm2d GP(RBF)	scat8_qm2d GP(RBF,ARD)	scat8_qm2d GP(Matern32)	scat8_qm2d GP(Matern32,ARD)	qm2d_coulomb GP(RBF)	qm2d_coulomb GP(RBF,ARD)	qm2d_coulomb GP(Matern32)	qm2d_coulomb GP(Matern32,ARD)	qm2d_cnn GP(RBF)	qm2d_cnn GP(RBF,ARD)	qm2d_cnn GP(Matern32)	qm2d_cnn GP(Matern32,ARD)
RMSE	0.4±0.02 (0 Nones, min: 0.36, max: 0.42)	0.31±0.09 (0 Nones, min: 0.19, max: 0.47)	0.42±0.05 (0 Nones, min: 0.34, max: 0.5)	0.32±0.08 (0 Nones, min: 0.24, max: 0.45)	0.39±0.06 (0 Nones, min: 0.32, max: 0.5)	**0.27±0.09 (0 Nones, min: 0.19, max: 0.45)**	0.42±0.04 (0 Nones, min: 0.37, max: 0.46)	0.29±0.03 (0 Nones, min: 0.25, max: 0.33)	0.57±0.11 (0 Nones, min: 0.38, max: 0.73)	0.58±0.09 (0 Nones, min: 0.52, max: 0.77)	0.53±0.1 (0 Nones, min: 0.39, max: 0.69)	0.46±0.03 (0 Nones, min: 0.42, max: 0.5)	0.53±0.04 (0 Nones, min: 0.47, max: 0.6)	0.63±0.07 (0 Nones, min: 0.54, max: 0.72)	0.53±0.04 (0 Nones, min: 0.47, max: 0.59)	0.59±0.06 (0 Nones, min: 0.51, max: 0.66)
RMSE (unnormalized)	71.85±1.34 (0 Nones, min: 69.65, max: 73.51)	54.74±11.34 (0 Nones, min: 44.55, max: 76.61)	75.14±5.64 (0 Nones, min: 68.7, max: 84.43)	57.22±6.4 (0 Nones, min: 51.22, max: 68.53)	71.58±6.62 (0 Nones, min: 65.55, max: 84.14)	**46.65±10.0 (0 Nones, min: 38.81, max: 66.16)**	75.49±2.43 (0 Nones, min: 70.75, max: 77.4)	49.93±3.56 (0 Nones, min: 43.83, max: 54.99)	105.44±13.32 (0 Nones, min: 83.61, max: 123.37)	101.37±9.84 (0 Nones, min: 95.37, max: 121.02)	95.14±11.07 (0 Nones, min: 78.17, max: 111.61)	84.4±7.78 (0 Nones, min: 75.23, max: 98.05)	97.52±4.54 (0 Nones, min: 92.07, max: 105.14)	108.4±6.73 (0 Nones, min: 99.12, max: 118.17)	95.92±3.36 (0 Nones, min: 90.95, max: 101.21)	106.6±5.88 (0 Nones, min: 97.22, max: 112.39)
MAE	0.29±0.02 (0 Nones, min: 0.27, max: 0.32)	0.21±0.05 (0 Nones, min: 0.15, max: 0.29)	0.32±0.03 (0 Nones, min: 0.26, max: 0.35)	0.22±0.05 (0 Nones, min: 0.16, max: 0.29)	0.3±0.05 (0 Nones, min: 0.24, max: 0.38)	**0.2±0.06 (0 Nones, min: 0.15, max: 0.31)**	0.32±0.03 (0 Nones, min: 0.28, max: 0.36)	0.2±0.02 (0 Nones, min: 0.18, max: 0.23)	0.41±0.09 (0 Nones, min: 0.27, max: 0.54)	0.41±0.07 (0 Nones, min: 0.33, max: 0.54)	0.34±0.05 (0 Nones, min: 0.27, max: 0.42)	0.3±0.02 (0 Nones, min: 0.28, max: 0.32)	0.41±0.03 (0 Nones, min: 0.37, max: 0.46)	0.48±0.05 (0 Nones, min: 0.41, max: 0.55)	0.41±0.03 (0 Nones, min: 0.37, max: 0.45)	0.46±0.05 (0 Nones, min: 0.4, max: 0.51)
MAE (unnormalized)	53.16±2.13 (0 Nones, min: 50.98, max: 56.63)	37.7±5.41 (0 Nones, min: 33.11, max: 47.23)	56.3±2.66 (0 Nones, min: 52.75, max: 59.36)	40.0±3.56 (0 Nones, min: 33.74, max: 44.63)	55.4±4.51 (0 Nones, min: 50.35, max: 63.45)	**34.74±5.78 (0 Nones, min: 30.55, max: 46.15)**	58.08±1.73 (0 Nones, min: 55.97, max: 60.48)	35.02±1.89 (0 Nones, min: 32.43, max: 37.02)	75.89±11.42 (0 Nones, min: 57.73, max: 91.45)	70.93±7.89 (0 Nones, min: 62.14, max: 85.35)	60.97±4.82 (0 Nones, min: 54.12, max: 68.44)	55.5±4.17 (0 Nones, min: 48.64, max: 60.34)	76.17±3.89 (0 Nones, min: 71.19, max: 83.08)	83.13±4.31 (0 Nones, min: 78.04, max: 89.47)	73.54±2.24 (0 Nones, min: 71.01, max: 77.49)	82.53±4.14 (0 Nones, min: 74.92, max: 86.41)
NLL (full)	0.01±0.08 (0 Nones, min: -0.08, max: 0.12)	5.0±2.48 (0 Nones, min: 0.51, max: 7.92)	0.01±0.11 (0 Nones, min: -0.18, max: 0.12)	0.4±0.21 (0 Nones, min: 0.03, max: 0.59)	-0.02±0.16 (0 Nones, min: -0.25, max: 0.18)	4.67±1.09 (0 Nones, min: 3.34, max: 6.61)	0.03±0.1 (0 Nones, min: -0.13, max: 0.17)	-0.22±0.05 (0 Nones, min: -0.28, max: -0.15)	0.34±0.3 (0 Nones, min: -0.11, max: 0.7)	0.34±0.21 (0 Nones, min: 0.05, max: 0.71)	-0.1±0.06 (0 Nones, min: -0.18, max: -0.02)	**-0.25±0.07 (0 Nones, min: -0.32, max: -0.14)**	0.4±0.12 (0 Nones, min: 0.27, max: 0.62)	2.63±0.75 (0 Nones, min: 1.53, max: 3.5)	0.42±0.07 (0 Nones, min: 0.29, max: 0.5)	0.75±0.13 (0 Nones, min: 0.58, max: 0.92)
NLL (diag)	321.84±61.11 (0 Nones, min: 235.48, max: 399.98)	3321.82±596.75 (0 Nones, min: 2385.3, max: 3983.21)	376.82±52.52 (0 Nones, min: 274.41, max: 422.53)	601.87±272.86 (0 Nones, min: 146.21, max: 997.56)	358.01±123.25 (0 Nones, min: 191.07, max: 533.57)	2605.99±829.87 (0 Nones, min: 1829.93, max: 4208.84	420.3±88.51 (0 Nones, min: 282.31, max: 516.29)	**188.63±93.07 (0 Nones, min: 57.95, max: 306.09)**	613.67±218.78 (0 Nones, min: 215.84, max: 865.11)	608.5±165.65 (0 Nones, min: 437.69, max: 919.98)	428.51±83.68 (0 Nones, min: 325.58, max: 566.25)	294.13±49.77 (0 Nones, min: 243.2, max: 386.98)	638.0±65.99 (0 Nones, min: 550.91, max: 753.6)	1542.08±269.73 (0 Nones, min: 1333.29, max: 20	652.33±70.14 (0 Nones, min: 547.01, max: 758.77)	880.21±120.41 (0 Nones, min: 743.54, max: 1041.24)
Time (s)	185.16±94.0 (0 Nones, min: 46.82, max: 314.97)	276.45±95.57 (0 Nones, min: 142.07, max: 414.31)	182.25±95.5 (0 Nones, min: 44.89, max: 317.22)	269.2±89.72 (0 Nones, min: 144.06, max: 395.97)	182.59±94.28 (0 Nones, min: 44.21, max: 310.47)	823.61±117.08 (0 Nones, min: 652.15, max: 989.55)	185.93±95.02 (0 Nones, min: 45.23, max: 313.09)	814.66±109.13 (0 Nones, min: 656.24, max: 967.6)	175.94±96.01 (0 Nones, min: 43.53, max: 320.73)	179.7±87.38 (0 Nones, min: 55.66, max: 302.76)	168.32±87.39 (0 Nones, min: 44.76, max: 291.05)	176.28±85.76 (0 Nones, min: 55.58, max: 297.07)	164.01±85.67 (0 Nones, min: 43.5, max: 286.5)	176.25±86.82 (0 Nones, min: 54.57, max: 299.38)	166.71±85.27 (0 Nones, min: 44.87, max: 286.28)	176.06±85.99 (0 Nones, min: 54.8, max: 297.4)

QM2D With Dropped Duplicates (**168** train (**0.04**), 841 test (0.2), overall: 4201 planar molecules). Core, Dirac, valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats

2025_02_03.ipynb	Scattering rotation-invariant				Vanilla Scattering				Coulomb				CNN			
	scat8_rotinv_qm2d GP(RBF)	scat8_rotinv_qm2d GP(RBF,ARD)	scat8_rotinv_qm2d GP(Matern32)	scat8_rotinv_qm2d GP(Matern32,ARD)	scat8_qm2d GP(RBF)	scat8_qm2d GP(RBF,ARD)	scat8_qm2d GP(Matern32)	scat8_qm2d GP(Matern32,ARD)	qm2d_coulomb GP(RBF)	qm2d_coulomb GP(RBF,ARD)	qm2d_coulomb GP(Matern32)	qm2d_coulomb GP(Matern32,ARD)	qm2d_cnn GP(RBF)	qm2d_cnn GP(RBF,ARD)	qm2d_cnn GP(Matern32)	qm2d_cnn GP(Matern32,ARD)
RMSE	0.33±0.02 (0 Nones, min: 0.29, max: 0.35)	0.25±0.07 (0 Nones, min: 0.15, max: 0.36)	0.39±0.05 (0 Nones, min: 0.32, max: 0.44)	0.22±0.02 (0 Nones, min: 0.18, max: 0.26)	0.39±0.05 (0 Nones, min: 0.34, max: 0.49)	**0.21±0.05 (0 Nones, min: 0.15, max: 0.3)**	0.4±0.03 (0 Nones, min: 0.36, max: 0.44)	0.19±0.01 (0 Nones, min: 0.18, max: 0.21)	0.48±0.09 (0 Nones, min: 0.39, max: 0.63)	0.41±0.03 (0 Nones, min: 0.37, max: 0.46)	0.54±0.02 (0 Nones, min: 0.52, max: 0.56)	0.42±0.08 (0 Nones, min: 0.34, max: 0.51)	0.51±0.04 (0 Nones, min: 0.46, max: 0.57)	0.62±0.06 (0 Nones, min: 0.57, max: 0.73)	0.51±0.03 (0 Nones, min: 0.48, max: 0.56)	0.48±0.04 (0 Nones, min: 0.43, max: 0.54)
RMSE (unnormalized)	62.87±3.34 (0 Nones, min: 57.19, max: 66.12)	42.44±8.3 (0 Nones, min: 27.43, max: 51.79)	70.8±5.61 (0 Nones, min: 62.97, max: 75.88)	42.15±2.98 (0 Nones, min: 38.1, max: 47.25)	68.0±5.54 (0 Nones, min: 61.72, max: 75.98)	**35.96±4.33 (0 Nones, min: 29.76, max: 42.27)**	69.4±4.45 (0 Nones, min: 64.54, max: 76.5)	37.08±2.06 (0 Nones, min: 34.9, max: 40.83)	87.46±8.8 (0 Nones, min: 75.82, max: 102.67)	77.71±5.81 (0 Nones, min: 71.97, max: 86.23)	95.05±2.81 (0 Nones, min: 90.42, max: 98.28)	75.93±7.83 (0 Nones, min: 68.08, max: 86.77)	90.55±4.14 (0 Nones, min: 84.82, max: 95.39)	107.15±7.41 (0 Nones, min: 97.91, max: 114.73)	90.85±5.67 (0 Nones, min: 84.71, max: 100.64)	91.3±3.95 (0 Nones, min: 85.63, max: 96.09)
MAE	0.24±0.01 (0 Nones, min: 0.22, max: 0.26)	0.16±0.03 (0 Nones, min: 0.12, max: 0.21)	0.29±0.03 (0 Nones, min: 0.25, max: 0.33)	0.15±0.01 (0 Nones, min: 0.13, max: 0.16)	0.29±0.04 (0 Nones, min: 0.26, max: 0.36)	**0.15±0.03 (0 Nones, min: 0.11, max: 0.19)**	0.3±0.02 (0 Nones, min: 0.28, max: 0.33)	0.14±0.01 (0 Nones, min: 0.12, max: 0.15)	0.31±0.06 (0 Nones, min: 0.26, max: 0.42)	0.25±0.02 (0 Nones, min: 0.23, max: 0.27)	0.33±0.01 (0 Nones, min: 0.32, max: 0.34)	0.27±0.03 (0 Nones, min: 0.23, max: 0.31)	0.39±0.03 (0 Nones, min: 0.35, max: 0.43)	0.47±0.04 (0 Nones, min: 0.43, max: 0.53)	0.39±0.02 (0 Nones, min: 0.37, max: 0.43)	0.37±0.03 (0 Nones, min: 0.34, max: 0.42)
MAE (unnormalized)	46.99±1.71 (0 Nones, min: 44.27, max: 49.0)	28.41±3.73 (0 Nones, min: 21.49, max: 31.96)	53.08±3.47 (0 Nones, min: 48.22, max: 56.99)	28.71±1.66 (0 Nones, min: 25.91, max: 30.83)	50.9±3.7 (0 Nones, min: 46.87, max: 56.77)	**25.52±1.72 (0 Nones, min: 23.0, max: 27.36)**	52.43±2.44 (0 Nones, min: 49.74, max: 56.04)	26.22±1.72 (0 Nones, min: 23.88, max: 28.12)	57.03±5.99 (0 Nones, min: 51.55, max: 68.12)	48.05±3.62 (0 Nones, min: 43.53, max: 53.48)	58.55±1.15 (0 Nones, min: 57.69, max: 60.78)	48.48±2.36 (0 Nones, min: 46.05, max: 51.35)	69.42±2.85 (0 Nones, min: 65.02, max: 72.31)	80.63±5.35 (0 Nones, min: 74.33, max: 88.02)	69.95±4.45 (0 Nones, min: 63.76, max: 77.0)	71.19±3.06 (0 Nones, min: 66.55, max: 74.51)
NLL (full)	-0.15±0.03 (0 Nones, min: -0.18, max: -0.11)	5.55±1.66 (0 Nones, min: 3.17, max: 8.11)	-0.03±0.1 (0 Nones, min: -0.16, max: 0.08)	-0.0±0.14 (0 Nones, min: -0.14, max: 0.24)	-0.09±0.13 (0 Nones, min: -0.21, max: 0.14)	3.03±0.57 (0 Nones, min: 2.45, max: 4.02)	0.05±0.06 (0 Nones, min: -0.02, max: 0.15)	**-0.62±0.1 (0 Nones, min: -0.73, max: -0.49)**	-0.02±0.24 (0 Nones, min: -0.23, max: 0.42)	-0.2±0.07 (0 Nones, min: -0.28, max: -0.12)	-0.16±0.02 (0 Nones, min: -0.19, max: -0.14)	-0.31±0.09 (0 Nones, min: -0.4, max: -0.2)	0.38±0.06 (0 Nones, min: 0.31, max: 0.46)	2.17±0.34 (0 Nones, min: 1.87, max: 2.8)	0.41±0.03 (0 Nones, min: 0.36, max: 0.45)	0.53±0.09 (0 Nones, min: 0.43, max: 0.65)
NLL (diag)	166.42±43.77 (0 Nones, min: 102.51, max: 239.06)	1981.49±396.12 (0 Nones, min: 1540.93, max: 2678.71	321.64±81.78 (0 Nones, min: 183.69, max: 418.79)	101.5±183.59 (0 Nones, min: -123.79, max: 307.33)	314.0±104.77 (0 Nones, min: 223.4, max: 515.16)	1440.59±168.3 (0 Nones, min: 1249.39, max: 1748.87)	388.93±56.31 (0 Nones, min: 300.29, max: 452.96)	**-204.86±96.41 (0 Nones, min: -344.6, max: -109.77)**	321.52±174.27 (0 Nones, min: 144.95, max: 606.81)	119.53±49.81 (0 Nones, min: 66.32, max: 186.07)	390.62±39.64 (0 Nones, min: 338.44, max: 453.09)	205.25±109.02 (0 Nones, min: 72.43, max: 358.68)	626.15±86.22 (0 Nones, min: 516.2, max: 762.28)	1470.67±166.24 (0 Nones, min: 1308.71, max: 17	600.89±43.28 (0 Nones, min: 544.53, max: 651.81)	697.84±138.75 (0 Nones, min: 519.8, max: 886.02)
Time (s)	185.24±93.42 (0 Nones, min: 46.74, max: 316.32)	275.14±93.43 (0 Nones, min: 144.61, max: 409.41)	202.66±110.32 (0 Nones, min: 46.08, max: 345.77)	271.11±92.19 (0 Nones, min: 144.81, max: 404.02)	186.62±91.9 (0 Nones, min: 49.94, max: 313.45)	821.53±111.82 (0 Nones, min: 657.53, max: 977.16)	185.18±92.88 (0 Nones, min: 47.01, max: 312.21)	835.49±108.88 (0 Nones, min: 667.53, max: 974.53)	185.61±105.98 (0 Nones, min: 48.04, max: 354.99)	181.29±88.78 (0 Nones, min: 56.49, max: 307.72)	170.54±88.21 (0 Nones, min: 45.47, max: 295.55)	178.42±85.97 (0 Nones, min: 56.74, max: 300.07)	164.84±84.55 (0 Nones, min: 44.37, max: 284.0)	178.35±87.52 (0 Nones, min: 55.19, max: 301.71)	168.8±87.15 (0 Nones, min: 45.21, max: 292.03)	177.9±86.04 (0 Nones, min: 55.94, max: 299.01)

QM2D With Dropped Duplicates (**210** train (**0.05**), 841 test (0.2), overall: 4201 planar molecules). Core, Dirac, valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats

2025_02_03.ipynb	Scattering rotation-invariant				Vanilla Scattering				Coulomb				CNN			
	scat8_rotinv_qm2d GP(RBF)	scat8_rotinv_qm2d GP(RBF,ARD)	scat8_rotinv_qm2d GP(Matern32)	scat8_rotinv_qm2d GP(Matern32,ARD)	scat8_qm2d GP(RBF)	scat8_qm2d GP(RBF,ARD)	scat8_qm2d GP(Matern32)	scat8_qm2d GP(Matern32,ARD)	qm2d_coulomb GP(RBF)	qm2d_coulomb GP(RBF,ARD)	qm2d_coulomb GP(Matern32)	qm2d_coulomb GP(Matern32,ARD)	qm2d_cnn GP(RBF)	qm2d_cnn GP(RBF,ARD)	qm2d_cnn GP(Matern32)	qm2d_cnn GP(Matern32,ARD)
RMSE	0.31±0.03 (0 Nones, min: 0.26, max: 0.33)	0.22±0.04 (0 Nones, min: 0.16, max: 0.29)	0.35±0.03 (0 Nones, min: 0.32, max: 0.39)	0.22±0.04 (0 Nones, min: 0.17, max: 0.3)	0.31±0.02 (0 Nones, min: 0.3, max: 0.36)	0.2±0.08 (0 Nones, min: 0.13, max: 0.35)	0.33±0.03 (0 Nones, min: 0.29, max: 0.36)	**0.18±0.01 (0 Nones, min: 0.16, max: 0.2)**	0.47±0.07 (0 Nones, min: 0.37, max: 0.56)	0.38±0.03 (0 Nones, min: 0.33, max: 0.42)	0.5±0.08 (0 Nones, min: 0.39, max: 0.61)	0.39±0.02 (0 Nones, min: 0.37, max: 0.42)	0.42±0.03 (0 Nones, min: 0.37, max: 0.46)	0.52±0.04 (0 Nones, min: 0.47, max: 0.58)	0.45±0.04 (0 Nones, min: 0.41, max: 0.51)	0.51±0.05 (0 Nones, min: 0.43, max: 0.56)
RMSE (unnormalized)	57.88±2.12 (0 Nones, min: 54.71, max: 61.1)	38.98±6.43 (0 Nones, min: 29.85, max: 48.1)	63.5±3.43 (0 Nones, min: 57.87, max: 68.42)	38.77±8.27 (0 Nones, min: 29.48, max: 54.27)	58.07±2.46 (0 Nones, min: 55.68, max: 62.6)	35.56±14.71 (0 Nones, min: 25.56, max: 64.69)	59.38±2.96 (0 Nones, min: 56.03, max: 63.77)	**32.44±1.87 (0 Nones, min: 30.51, max: 35.6)**	88.1±11.07 (0 Nones, min: 71.6, max: 100.84)	71.46±4.98 (0 Nones, min: 65.12, max: 78.52)	87.3±9.89 (0 Nones, min: 77.12, max: 103.33)	71.76±2.0 (0 Nones, min: 69.82, max: 74.86)	80.2±2.95 (0 Nones, min: 74.71, max: 82.97)	96.23±4.38 (0 Nones, min: 90.36, max: 101.37)	82.56±2.3 (0 Nones, min: 79.53, max: 86.5)	88.85±4.8 (0 Nones, min: 80.45, max: 94.42)
MAE	0.23±0.02 (0 Nones, min: 0.2, max: 0.25)	0.14±0.02 (0 Nones, min: 0.12, max: 0.17)	0.26±0.01 (0 Nones, min: 0.25, max: 0.27)	0.14±0.01 (0 Nones, min: 0.13, max: 0.16)	0.24±0.01 (0 Nones, min: 0.23, max: 0.26)	**0.12±0.01 (0 Nones, min: 0.1, max: 0.14)**	0.25±0.03 (0 Nones, min: 0.22, max: 0.28)	0.13±0.01 (0 Nones, min: 0.12, max: 0.13)	0.28±0.03 (0 Nones, min: 0.24, max: 0.31)	0.25±0.01 (0 Nones, min: 0.23, max: 0.27)	0.29±0.03 (0 Nones, min: 0.26, max: 0.33)	0.24±0.01 (0 Nones, min: 0.23, max: 0.24)	0.32±0.02 (0 Nones, min: 0.29, max: 0.35)	0.39±0.03 (0 Nones, min: 0.36, max: 0.44)	0.35±0.02 (0 Nones, min: 0.32, max: 0.38)	0.39±0.04 (0 Nones, min: 0.33, max: 0.43)
MAE (unnormalized)	43.47±1.21 (0 Nones, min: 42.32, max: 45.32)	24.27±1.99 (0 Nones, min: 21.89, max: 27.71)	47.02±0.87 (0 Nones, min: 45.61, max: 48.0)	25.23±2.2 (0 Nones, min: 21.93, max: 28.5)	43.78±0.76 (0 Nones, min: 42.92, max: 44.91)	**21.98±2.17 (0 Nones, min: 19.27, max: 25.07)**	45.75±2.27 (0 Nones, min: 42.56, max: 48.26)	22.83±0.77 (0 Nones, min: 21.75, max: 23.67)	52.92±4.1 (0 Nones, min: 46.75, max: 58.98)	46.33±2.35 (0 Nones, min: 43.93, max: 50.69)	50.9±2.51 (0 Nones, min: 48.06, max: 54.92)	43.26±1.97 (0 Nones, min: 40.35, max: 45.91)	61.86±1.63 (0 Nones, min: 59.52, max: 64.28)	73.25±3.22 (0 Nones, min: 68.61, max: 78.65)	64.12±0.97 (0 Nones, min: 62.98, max: 65.31)	68.15±3.82 (0 Nones, min: 62.74, max: 73.13)
NLL (full)	-0.14±0.07 (0 Nones, min: -0.24, max: -0.04)	3.15±0.39 (0 Nones, min: 2.58, max: 3.72)	-0.1±0.05 (0 Nones, min: -0.19, max: -0.05)	-0.35±0.12 (0 Nones, min: -0.49, max: -0.19)	-0.17±0.06 (0 Nones, min: -0.25, max: -0.09)	2.62±0.33 (0 Nones, min: 2.12, max: 3.05)	-0.09±0.08 (0 Nones, min: -0.2, max: 0.01)	**-0.7±0.07 (0 Nones, min: -0.78, max: -0.61)**	-0.16±0.07 (0 Nones, min: -0.25, max: -0.05)	-0.16±0.08 (0 Nones, min: -0.29, max: -0.08)	-0.2±0.08 (0 Nones, min: -0.31, max: -0.11)	-0.39±0.04 (0 Nones, min: -0.44, max: -0.32)	0.26±0.07 (0 Nones, min: 0.17, max: 0.34)	1.67±0.43 (0 Nones, min: 1.3, max: 2.49)	0.33±0.09 (0 Nones, min: 0.24, max: 0.48)	0.62±0.11 (0 Nones, min: 0.44, max: 0.8)
NLL (diag)	120.55±60.96 (0 Nones, min: 8.68, max: 195.26)	1668.3±435.11 (0 Nones, min: 1184.7, max: 2257.33)	223.82±29.54 (0 Nones, min: 178.13, max: 266.72)	-83.5±112.11 (0 Nones, min: -258.91, max: 91.26)	135.17±30.0 (0 Nones, min: 101.98, max: 185.74)	1309.9±291.25 (0 Nones, min: 967.12, max: 1657.69)	224.17±95.49 (0 Nones, min: 95.7, max: 324.82)	**-320.46±73.94 (0 Nones, min: -441.29, max: -257.85)**	180.27±46.75 (0 Nones, min: 105.55, max: 242.34)	132.83±61.44 (0 Nones, min: 65.65, max: 222.1)	249.55±74.93 (0 Nones, min: 169.19, max: 357.53)	84.81±29.3 (0 Nones, min: 35.09, max: 122.03)	426.03±60.71 (0 Nones, min: 342.47, max: 491.53	1131.79±169.63 (0 Nones, min: 1000.8, max: 146	508.59±63.77 (0 Nones, min: 440.08, max: 593.84)	716.13±134.99 (0 Nones, min: 515.68, max: 937.13)
Time (s)	185.54±94.8 (0 Nones, min: 46.16, max: 317.18)	277.73±90.8 (0 Nones, min: 150.09, max: 405.32)	214.67±100.56 (0 Nones, min: 70.3, max: 350.94)	274.79±91.94 (0 Nones, min: 145.8, max: 406.59)	184.08±95.06 (0 Nones, min: 46.16, max: 317.77)	818.61±110.13 (0 Nones, min: 661.17, max: 976.71)	184.01±93.03 (0 Nones, min: 48.19, max: 311.26)	813.73±109.49 (0 Nones, min: 649.42, max: 960.4)	193.93±91.81 (0 Nones, min: 65.22, max: 325.41)	182.03±87.67 (0 Nones, min: 57.08, max: 305.28)	172.71±88.33 (0 Nones, min: 46.54, max: 296.74)	180.23±86.47 (0 Nones, min: 57.89, max: 302.13)	165.69±85.12 (0 Nones, min: 44.92, max: 286.83)	178.47±86.9 (0 Nones, min: 55.73, max: 302.08)	169.49±86.86 (0 Nones, min: 45.52, max: 292.08)	178.5±85.64 (0 Nones, min: 57.45, max: 299.42)

QM2D With Dropped Duplicates (**420** train (**0.1**), 841 test (0.2), overall: 4201 planar molecules). Core, Dirac, valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats

2025_02_03.ipynb	Scattering rotation-invariant				Vanilla Scattering				Coulomb				CNN			
	scat8_rotinv_qm2d GP(RBF)	scat8_rotinv_qm2d GP(RBF,ARD)	scat8_rotinv_qm2d GP(Matern32)	scat8_rotinv_qm2d GP(Matern32,ARD)	scat8_qm2d GP(RBF)	scat8_qm2d GP(RBF,ARD)	scat8_qm2d GP(Matern32)	scat8_qm2d GP(Matern32,ARD)	qm2d_coulomb GP(RBF)	qm2d_coulomb GP(RBF,ARD)	qm2d_coulomb GP(Matern32)	qm2d_coulomb GP(Matern32,ARD)	qm2d_cnn GP(RBF)	qm2d_cnn GP(RBF,ARD)	qm2d_cnn GP(Matern32)	qm2d_cnn GP(Matern32,ARD)
RMSE	0.24±0.02 (0 Nones, min: 0.21, max: 0.28)	0.13±0.02 (0 Nones, min: 0.11, max: 0.16)	0.26±0.01 (0 Nones, min: 0.24, max: 0.28)	0.14±0.03 (0 Nones, min: 0.11, max: 0.21)	0.25±0.02 (0 Nones, min: 0.21, max: 0.27)	0.12±0.03 (0 Nones, min: 0.1, max: 0.19)	0.26±0.02 (0 Nones, min: 0.24, max: 0.3)	**0.12±0.02 (0 Nones, min: 0.11, max: 0.16)**	0.38±0.05 (0 Nones, min: 0.32, max: 0.47)	0.37±0.03 (0 Nones, min: 0.33, max: 0.42)	0.31±0.02 (0 Nones, min: 0.29, max: 0.34)	0.31±0.04 (0 Nones, min: 0.26, max: 0.36)	0.36±0.03 (0 Nones, min: 0.32, max: 0.39)	0.41±0.03 (0 Nones, min: 0.36, max: 0.45)	0.37±0.01 (0 Nones, min: 0.36, max: 0.39)	0.38±0.02 (0 Nones, min: 0.37, max: 0.41)
RMSE (unnormalized)	42.99±2.56 (0 Nones, min: 40.47, max: 47.85)	23.22±3.71 (0 Nones, min: 20.01, max: 28.65)	47.43±2.42 (0 Nones, min: 43.59, max: 50.22)	24.53±4.83 (0 Nones, min: 20.87, max: 33.81)	43.6±1.97 (0 Nones, min: 40.05, max: 45.81)	22.06±5.71 (0 Nones, min: 18.0, max: 33.31)	46.84±2.49 (0 Nones, min: 44.77, max: 51.65)	**21.81±3.24 (0 Nones, min: 19.83, max: 28.26)**	67.28±7.53 (0 Nones, min: 58.72, max: 79.1)	66.08±5.56 (0 Nones, min: 58.45, max: 74.21)	57.52±3.19 (0 Nones, min: 54.6, max: 63.58)	55.43±6.45 (0 Nones, min: 48.12, max: 63.34)	65.29±1.49 (0 Nones, min: 62.66, max: 66.65)	75.06±5.29 (0 Nones, min: 67.36, max: 81.08)	67.14±1.18 (0 Nones, min: 64.91, max: 68.12)	70.09±1.08 (0 Nones, min: 68.73, max: 71.36)
MAE	0.18±0.02 (0 Nones, min: 0.15, max: 0.2)	0.08±0.0 (0 Nones, min: 0.08, max: 0.09)	0.2±0.01 (0 Nones, min: 0.19, max: 0.21)	0.09±0.01 (0 Nones, min: 0.08, max: 0.1)	0.19±0.01 (0 Nones, min: 0.16, max: 0.21)	**0.08±0.0 (0 Nones, min: 0.08, max: 0.09)**	0.19±0.01 (0 Nones, min: 0.18, max: 0.21)	0.08±0.01 (0 Nones, min: 0.07, max: 0.1)	0.22±0.02 (0 Nones, min: 0.19, max: 0.24)	0.21±0.01 (0 Nones, min: 0.19, max: 0.23)	0.17±0.01 (0 Nones, min: 0.16, max: 0.17)	0.17±0.02 (0 Nones, min: 0.14, max: 0.19)	0.28±0.02 (0 Nones, min: 0.25, max: 0.3)	0.31±0.02 (0 Nones, min: 0.28, max: 0.34)	0.29±0.01 (0 Nones, min: 0.27, max: 0.3)	0.3±0.02 (0 Nones, min: 0.28, max: 0.32)
MAE (unnormalized)	31.42±1.99 (0 Nones, min: 29.65, max: 35.11)	15.19±0.35 (0 Nones, min: 14.74, max: 15.8)	35.82±0.98 (0 Nones, min: 34.1, max: 37.15)	15.79±0.72 (0 Nones, min: 15.14, max: 17.14)	33.0±1.44 (0 Nones, min: 30.52, max: 34.51)	**14.56±0.78 (0 Nones, min: 13.77, max: 15.86)**	35.09±0.75 (0 Nones, min: 33.93, max: 36.19)	15.5±0.87 (0 Nones, min: 13.98, max: 16.65)	38.57±2.5 (0 Nones, min: 34.53, max: 41.45)	36.89±2.32 (0 Nones, min: 34.05, max: 40.76)	31.6±1.12 (0 Nones, min: 29.99, max: 32.89)	29.73±1.97 (0 Nones, min: 27.32, max: 32.97)	50.4±1.0 (0 Nones, min: 49.17, max: 51.56)	57.18±3.45 (0 Nones, min: 52.22, max: 61.15)	51.51±0.4 (0 Nones, min: 50.9, max: 51.99)	54.48±1.47 (0 Nones, min: 51.96, max: 56.06)
NLL (full)	-0.4±0.09 (0 Nones, min: -0.53, max: -0.25)	-0.4±0.49 (0 Nones, min: -0.95, max: 0.38)	-0.28±0.03 (0 Nones, min: -0.34, max: -0.25)	-0.94±0.05 (0 Nones, min: -0.99, max: -0.87)	-0.33±0.05 (0 Nones, min: -0.41, max: -0.26)	1.19±0.42 (0 Nones, min: 0.36, max: 1.5)	-0.25±0.07 (0 Nones, min: -0.37, max: -0.18)	**-1.05±0.06 (0 Nones, min: -1.12, max: -0.96)**	-0.36±0.05 (0 Nones, min: -0.42, max: -0.3)	-0.24±0.05 (0 Nones, min: -0.31, max: -0.16)	-0.51±0.06 (0 Nones, min: -0.57, max: -0.41)	-0.61±0.06 (0 Nones, min: -0.7, max: -0.54)	0.21±0.06 (0 Nones, min: 0.14, max: 0.29)	0.79±0.22 (0 Nones, min: 0.46, max: 1.09)	0.24±0.03 (0 Nones, min: 0.2, max: 0.29)	0.36±0.06 (0 Nones, min: 0.29, max: 0.44)
NLL (diag)	-126.81±80.75 (0 Nones, min: -241.49, max: -6.57)	-108.4±416.62 (0 Nones, min: -511.1, max: 647.44)	-16.96±23.67 (0 Nones, min: -47.71, max: 17.42)	-610.74±39.59 (0 Nones, min: -656.7, max: -547.51)	-43.91±68.27 (0 Nones, min: -152.96, max: 54.86)	747.98±143.77 (0 Nones, min: 470.03, max: 885.45)	-4.77±40.57 (0 Nones, min: -69.0, max: 40.0)	**-667.39±80.04 (0 Nones, min: -804.38, max: -580.44)**	-31.58±39.87 (0 Nones, min: -90.41, max: 34.77)	-35.07±55.56 (0 Nones, min: -83.55, max: 60.72)	-179.45±27.48 (0 Nones, min: -217.91, max: -136.43)	-249.12±57.42 (0 Nones, min: -333.51, max: -174.78)	311.97±62.43 (0 Nones, min: 223.05, max: 386.88)	683.14±80.18 (0 Nones, min: 577.53, max: 824.08	341.84±35.15 (0 Nones, min: 279.42, max: 381.54)	438.84±46.34 (0 Nones, min: 376.19, max: 503.71)
Time (s)	196.24±98.24 (0 Nones, min: 45.38, max: 324.49)	287.78±96.3 (0 Nones, min: 145.27, max: 419.91)	222.65±97.82 (0 Nones, min: 84.24, max: 363.13)	288.16±97.39 (0 Nones, min: 146.93, max: 421.37)	185.89±91.05 (0 Nones, min: 49.33, max: 304.8)	795.28±127.84 (0 Nones, min: 610.54, max: 970.28)	186.92±92.74 (0 Nones, min: 45.36, max: 307.51)	791.04±121.11 (0 Nones, min: 620.56, max: 960.61)	203.39±95.27 (0 Nones, min: 68.93, max: 337.01)	194.16±96.55 (0 Nones, min: 60.19, max: 334.23)	178.19±89.22 (0 Nones, min: 48.49, max: 302.06)	185.64±87.41 (0 Nones, min: 59.92, max: 307.44)	172.12±87.08 (0 Nones, min: 45.95, max: 293.88)	184.84±87.37 (0 Nones, min: 58.79, max: 306.88)	175.67±88.32 (0 Nones, min: 47.39, max: 299.0)	185.68±87.81 (0 Nones, min: 59.34, max: 309.29)

QM2D With Dropped Duplicates (**630** train (**0.15**), 841 test (0.2), overall: 4201 planar molecules). Core, Dirac, valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats

2025_02_03.ipynb	Scattering rotation-invariant				Vanilla Scattering				Coulomb				CNN			
	scat8_rotinv_qm2d GP(RBF)	scat8_rotinv_qm2d GP(RBF,ARD)	scat8_rotinv_qm2d GP(Matern32)	scat8_rotinv_qm2d GP(Matern32,ARD)	scat8_qm2d GP(RBF)	scat8_qm2d GP(RBF,ARD)	scat8_qm2d GP(Matern32)	scat8_qm2d GP(Matern32,ARD)	qm2d_coulomb GP(RBF)	qm2d_coulomb GP(RBF,ARD)	qm2d_coulomb GP(Matern32)	qm2d_coulomb GP(Matern32,ARD)	qm2d_cnn GP(RBF)	qm2d_cnn GP(RBF,ARD)	qm2d_cnn GP(Matern32)	qm2d_cnn GP(Matern32,ARD)
RMSE	0.19±0.01 (0 Nones, min: 0.16, max: 0.2)	0.09±0.01 (0 Nones, min: 0.08, max: 0.1)	0.22±0.01 (0 Nones, min: 0.2, max: 0.23)	0.1±0.01 (0 Nones, min: 0.09, max: 0.11)	0.18±0.0 (0 Nones, min: 0.17, max: 0.18)	**0.08±0.0 (0 Nones, min: 0.08, max: 0.09)**	0.22±0.01 (0 Nones, min: 0.2, max: 0.23)	0.09±0.0 (0 Nones, min: 0.08, max: 0.1)	0.31±0.04 (0 Nones, min: 0.28, max: 0.39)	0.26±0.02 (0 Nones, min: 0.23, max: 0.29)	0.28±0.03 (0 Nones, min: 0.24, max: 0.31)	0.23±0.01 (0 Nones, min: 0.22, max: 0.24)	0.32±0.02 (0 Nones, min: 0.3, max: 0.36)	0.36±0.02 (0 Nones, min: 0.32, max: 0.39)	0.33±0.01 (0 Nones, min: 0.32, max: 0.34)	0.33±0.01 (0 Nones, min: 0.32, max: 0.35)
RMSE (unnormalized)	34.46±1.62 (0 Nones, min: 31.81, max: 36.24)	16.42±1.73 (0 Nones, min: 13.98, max: 18.89)	39.52±2.28 (0 Nones, min: 37.25, max: 42.57)	18.3±1.6 (0 Nones, min: 16.87, max: 21.15)	33.24±0.71 (0 Nones, min: 32.09, max: 34.05)	**15.42±0.47 (0 Nones, min: 15.09, max: 16.35)**	39.06±1.4 (0 Nones, min: 37.15, max: 40.8)	16.28±0.74 (0 Nones, min: 15.5, max: 17.61)	56.0±8.14 (0 Nones, min: 48.96, max: 70.99)	49.07±4.29 (0 Nones, min: 41.11, max: 52.49)	51.69±5.09 (0 Nones, min: 44.09, max: 59.32)	41.87±2.3 (0 Nones, min: 40.1, max: 46.29)	58.23±2.28 (0 Nones, min: 56.28, max: 62.61)	64.18±3.47 (0 Nones, min: 59.09, max: 69.47)	59.98±0.56 (0 Nones, min: 59.37, max: 60.88)	62.61±1.39 (0 Nones, min: 60.81, max: 64.84)
MAE	0.13±0.01 (0 Nones, min: 0.12, max: 0.14)	0.06±0.0 (0 Nones, min: 0.06, max: 0.07)	0.15±0.0 (0 Nones, min: 0.15, max: 0.16)	0.07±0.0 (0 Nones, min: 0.06, max: 0.07)	0.13±0.0 (0 Nones, min: 0.12, max: 0.14)	**0.06±0.0 (0 Nones, min: 0.06, max: 0.07)**	0.17±0.01 (0 Nones, min: 0.16, max: 0.17)	0.06±0.0 (0 Nones, min: 0.06, max: 0.07)	0.15±0.01 (0 Nones, min: 0.15, max: 0.17)	0.16±0.01 (0 Nones, min: 0.15, max: 0.17)	0.15±0.01 (0 Nones, min: 0.13, max: 0.16)	0.13±0.01 (0 Nones, min: 0.11, max: 0.14)	0.25±0.01 (0 Nones, min: 0.23, max: 0.28)	0.27±0.01 (0 Nones, min: 0.25, max: 0.29)	0.25±0.01 (0 Nones, min: 0.24, max: 0.26)	0.26±0.01 (0 Nones, min: 0.24, max: 0.27)
MAE (unnormalized)	23.8±1.0 (0 Nones, min: 22.79, max: 25.66)	11.56±0.58 (0 Nones, min: 10.63, max: 12.35)	28.34±0.42 (0 Nones, min: 27.85, max: 29.05)	12.46±0.43 (0 Nones, min: 11.84, max: 13.07)	24.33±0.88 (0 Nones, min: 23.19, max: 25.75)	**11.51±0.11 (0 Nones, min: 11.4, max: 11.7)**	29.98±1.09 (0 Nones, min: 28.6, max: 31.6)	11.89±0.16 (0 Nones, min: 11.64, max: 12.11)	27.87±2.2 (0 Nones, min: 24.82, max: 30.52)	29.89±1.36 (0 Nones, min: 27.22, max: 31.07)	27.15±1.54 (0 Nones, min: 25.33, max: 29.37)	23.12±1.61 (0 Nones, min: 20.52, max: 25.57)	45.57±1.41 (0 Nones, min: 43.95, max: 48.18)	48.71±1.44 (0 Nones, min: 46.27, max: 50.06)	45.99±0.5 (0 Nones, min: 45.27, max: 46.66)	48.5±1.09 (0 Nones, min: 46.79, max: 49.93)
NLL (full)	-0.69±0.05 (0 Nones, min: -0.79, max: -0.64)	-1.04±0.13 (0 Nones, min: -1.16, max: -0.79)	-0.49±0.02 (0 Nones, min: -0.51, max: -0.47)	-1.14±0.07 (0 Nones, min: -1.2, max: -1.01)	-0.67±0.06 (0 Nones, min: -0.75, max: -0.59)	-0.06±0.52 (0 Nones, min: -1.05, max: 0.48)	-0.39±0.03 (0 Nones, min: -0.44, max: -0.36)	**-1.26±0.05 (0 Nones, min: -1.32, max: -1.2)**	-0.68±0.08 (0 Nones, min: -0.79, max: -0.59)	-0.48±0.05 (0 Nones, min: -0.56, max: -0.42)	-0.66±0.06 (0 Nones, min: -0.74, max: -0.59)	-0.78±0.05 (0 Nones, min: -0.84, max: -0.69)	0.15±0.03 (0 Nones, min: 0.12, max: 0.21)	0.26±0.06 (0 Nones, min: 0.14, max: 0.31)	0.17±0.03 (0 Nones, min: 0.11, max: 0.19)	0.21±0.04 (0 Nones, min: 0.16, max: 0.27)
NLL (diag)	-404.25±52.37 (0 Nones, min: -491.04, max: -336.88)	-732.46±102.01 (0 Nones, min: -826.31, max: -539.95)	-214.73±24.82 (0 Nones, min: -254.85, max: -182.96)	-826.37±75.6 (0 Nones, min: -902.47, max: -688.28)	-373.77±36.02 (0 Nones, min: -416.1, max: -328.38)	280.87±33.74 (0 Nones, min: 232.86, max: 334.2)	-151.05±26.16 (0 Nones, min: -194.24, max: -124.87)	**-948.49±20.8 (0 Nones, min: -974.07, max: -913.56)**	-379.79±57.21 (0 Nones, min: -440.32, max: -286.04)	-242.45±29.26 (0 Nones, min: -277.97, max: -204.59)	-330.09±65.84 (0 Nones, min: -417.57, max: -224.23)	-457.33±49.75 (0 Nones, min: -537.43, max: -386.77)	213.09±41.69 (0 Nones, min: 176.21, max: 290.9)	331.93±49.78 (0 Nones, min: 234.52, max: 372.11	240.19±27.12 (0 Nones, min: 188.24, max: 267.56)	273.92±44.89 (0 Nones, min: 227.16, max: 346.77)
Time (s)	189.09±95.62 (0 Nones, min: 45.43, max: 314.75)	286.27±98.25 (0 Nones, min: 146.86, max: 425.77)	231.38±99.76 (0 Nones, min: 84.58, max: 370.47)	287.56±98.97 (0 Nones, min: 145.73, max: 429.43)	182.92±91.91 (0 Nones, min: 46.76, max: 311.95)	788.24±113.95 (0 Nones, min: 624.58, max: 948.27)	191.74±96.72 (0 Nones, min: 49.53, max: 328.29)	776.3±106.72 (0 Nones, min: 628.18, max: 933.76)	221.94±94.69 (0 Nones, min: 79.91, max: 350.69)	204.07±93.91 (0 Nones, min: 67.42, max: 335.32)	184.7±92.28 (0 Nones, min: 47.17, max: 310.67)	191.4±89.3 (0 Nones, min: 59.31, max: 314.52)	179.0±90.23 (0 Nones, min: 45.67, max: 302.61)	191.24±89.82 (0 Nones, min: 59.8, max: 316.75)	182.86±90.89 (0 Nones, min: 47.68, max: 307.24)	193.05±90.99 (0 Nones, min: 58.79, max: 319.01)